

# Ultrasound for Venous Local Pulse Wave Velocity: Comparison of Pulse Transit Time Methods

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**Abstract**— Venous pulse wave velocity (vPWV) is a potential marker for determining the state of venous hemodynamics, venosclerosis, and vascular filling. Although there have been several studies on pulse wave velocity through blood vessels, the majority have focused on arteries, with only limited studies on veins. To our knowledge, this study is the first to compare the local vPWV estimation metrics. An in vivo study was conducted on 10 participants where the jugular venous pulses (JVP) from two proximal sites were simultaneously acquired using a dual-element high frame rate system. The local vPWV was computed using different transit time-based techniques. The study demonstrates the comparison between vPWV ranges computed using thresholding, fiduciary point (c and v) and correlation-based approaches indicated as  $vPWV|_{th}$ ,  $vPWV|_c$ ,  $vPWV|_v$  and  $vPWV|_{Xcorr}$  respectively. High fidelity echo frames were acquired from the jugular vein (JV) at a temporal resolution of 2 ms and an amplitude resolution of 10  $\mu$ m. The study findings indicated that the vPWV computed using various transit time metrics were comparable without significant bias ( $p > 0.05$ ). Among the vPWV metrics,  $vPWV|_{th}$  had the lowest beat-to-beat variation (CoV = 18 %). The mean deviations in  $vPWV|_c$ ,  $vPWV|_v$  and  $vPWV|_{Xcorr}$  values from  $vPWV|_{th}$  were 0.28, 0.17 and 0.22 m/s respectively, where the average beat-to-beat variation was minimal. The results suggested that the thresholding and cross-correlation metrics offered better performance in comparison with the fiduciary point techniques for vPWV estimation.

**Clinical Relevance**— The study demonstrated the potential of direct transit time methods to reliably estimate the local vPWV on the internal jugular vein.

## I. INTRODUCTION

Pulse wave velocity (PWV) is an established marker to estimate the stiffness and associated hemodynamic changes in blood vessels. Due to inherent cardiac pulsatility and increased blood pressure in arteries a deluge of studies has been reported on arterial PWV measurements and their association with

other vascular markers. The vPWV has gained more attention lately. It has a good sensitivity to blood volume changes and vascular filling [1]. Few studies reported vPWV as a surrogate marker to estimate the central venous pressure [2], obviating the need for invasive measurements. Given the non-pulsatile nature of veins, the vPWV measurements are quite challenging thus only limited research is predicated on veins. The venous pulsations are primarily due to rhythmic motion of the vessel smooth muscles and the action of skeletal muscles during body movements [3].

The conventional vPWV assessments relied on monitoring the transit time difference between venous pulses from two known locations captured using pressure catheters [4]. An indirect approach to estimate vPWV is by inducing pressure pulses at limb extremities by a pneumatic compression and detecting the time shift at the basilic vein [5]. While these indirect methods employ active measurement systems to monitor the vPWV from peripheral veins, none utilized the inherent venous pulsations [2], [5], [6]. The internal jugular veins (IJV) provide visible pulsations in the neck as it drains deoxygenated blood from the head back to the heart. The JVP reflects variations in atrial pressure because of its direct connection with the right atrium. Due to the pulsatile nature of IJV, the existing arterial pulse wave velocity assessment techniques are implementable on the JVP signals. Indirect arterial PWV assessment methods include biomechanical equations that rely on the measurement of pressure and flow at a single site [7]. IJV pressure measurements are trivial due to their feeble values and hence the vPWV estimation using biomechanical equations isn't straight forward.

The simplest direct method for PWV estimation is based on the pulse transit time (PTT) principle. PTT is the time taken by a blood pulse to propagate between two sites located in the vessel [7]. The PTT estimation techniques in arteries are based on either shift in fiduciary point (waveform foot, peak, first derivative maximum, dirotic notch etc.) locations, or statistical methods (phase offset, correlation) [8]. Even though numerous PTT based arterial PWV measurements are reported, none have been on the IJV to the best of our knowledge. This study investigates the viability of transit time-based approaches to estimate vPWV and quantitatively compares them to evaluate the optimum approach.

We have demonstrated and validated the basic principle of recording high fidelity JVP signals (temporal resolution = 4

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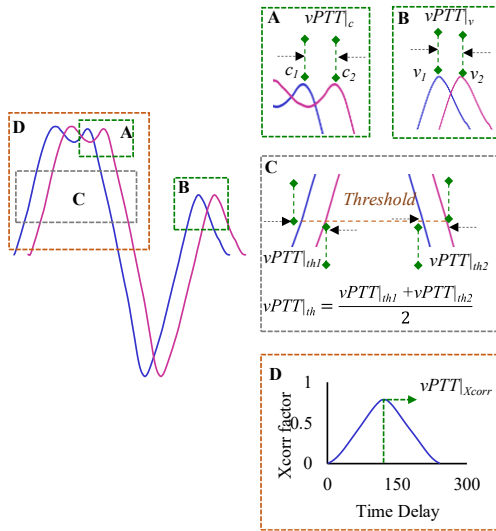


Fig 1. Representative illustration of different transit time based vPWV techniques where: A- fiduciary point c, B-fiduciary point v, C- thresholding and D- Cross-correlation.

ms) in our earlier work using a single channel image free ultrasound system [8]. We have augmented and advanced it to a dual element system operating at a frame rate of 500 Hz to evaluate the local vPWV. The measurement system acquires JVP signals simultaneously from two sites along the jugular vein and applies different transit time approaches to evaluate the local vPWV. The study and instrumentation details are elaborated in section II, followed by the comparative results of various local vPWV metrics discussed in Section III.

## II. MATERIALS AND METHODS

### A. Study Objectives

The study objectives are:

- To evaluate the feasibility of simultaneous dual JVP signal acquisition from a close proximity (35 mm apart).
- To assess the viability of vPWV calculation using the transit time-based approaches:

- 1) Fiduciary point c of JVP ( $vPWV|_c$ )
  - 2) Fiduciary point v of JVP ( $vPWV|_v$ )
  - 3) Thresholding ( $vPWV|_{th}$ )
  - 4) Cross- Correlation ( $vPWV|_{Xcorr}$ )
- c) To determine the optimum fit for vPWV computation by comparing different PTT techniques.

### B. Participants

The study was conducted on a cohort of 10 healthy participants aged between 20 to 30 years. The selected population was devoid of venous or cardiovascular diseases. The in-vivo measurements were conducted in accordance with the protocols (approved by the Institutional Ethical Committee, IIT Madras, IEC/2021- 01/JJ/07) that were compliant with the Helsinki Declaration 1975, amended in 2013. Prior to the data collection, written consent was obtained from all participants.

### C. Study Protocol

The anthropometric measurements, height weight and body mass index (BMI) of the participant were recorded. The participant was requested to rest in supine posture for 5 minutes. The blood pressure measurements were taken once the subject is relaxed. The developed dual ultrasound system was used to simultaneously acquire vessel wall echoes from the two proximally spaced sites along the IJV. A LabVIEW (LabVIEW 2019, 32-bit, National Instruments Co., United States) based software application was used to acquire and process the JVP signals. All the measurements were carried out in a controlled laboratory environment at an ambient temperature of 22 – 24 °C.

### D. Data Collection and Processing

Owing to the unique morphological characteristics of JVP signals, a high frame rate acquisition is necessary to preserve the pulse characteristics [10]. The developed measurement system could acquire high fidelity JVP signals from the two proximally located IJV locations at a frame rate of 500 Hz. The acquisition system consists of a dual element ultrasound probe with transducers (center frequency = 10 MHz and diameter =

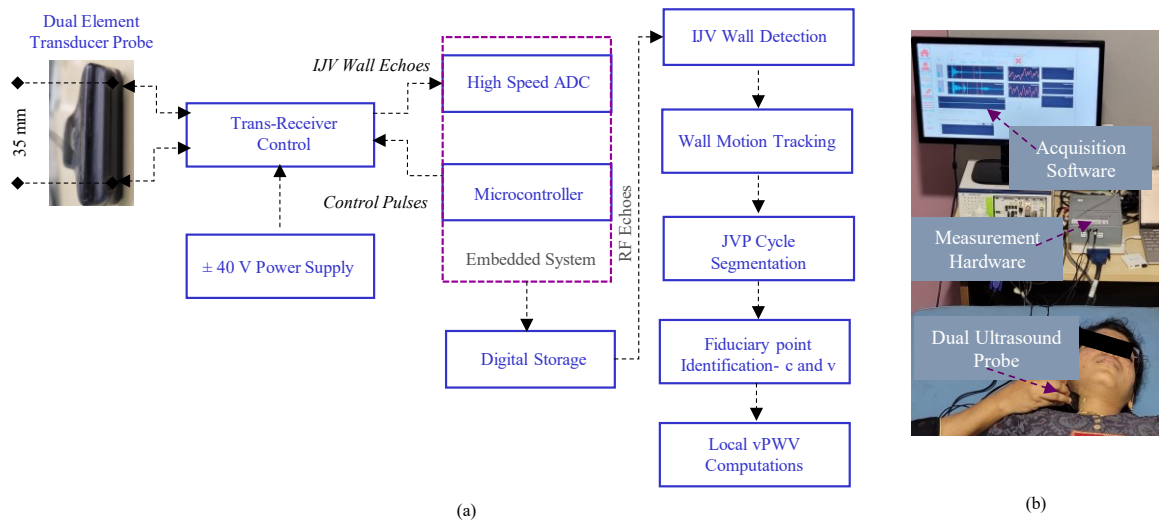


Fig 2. (a) Image free ultrasound system architecture for vPWV measurements (b) Illustration of vPWV measurement system using dual ultrasound probe on the internal jugular vein

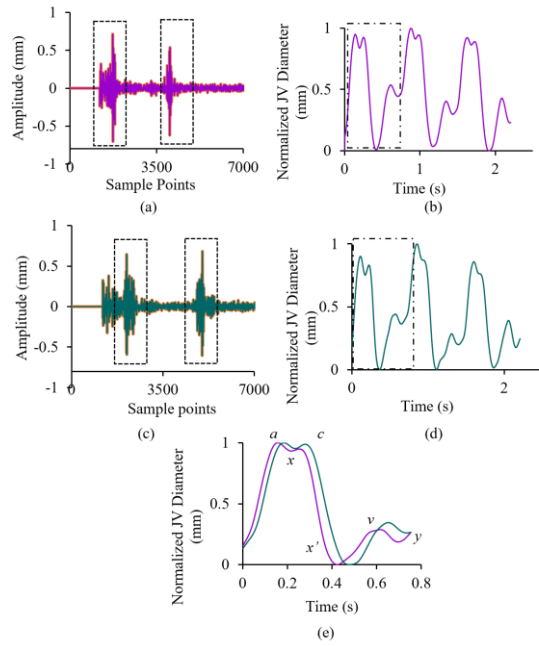


Fig 3. (a) RF echo frames obtained from proximal transducer (b) RF echo frames obtained from distal transducer (c) JVP cycles by tracking proximal echoes (d) JVP cycles by tracking distal echoes (e) Overplot of the segmented JVP cycles from the distal and proximal sites.

5 mm) operating in pulse echo modality spaced at center-to-center distance of 35 mm.

The pulser-receiver unit transmits and receives echo pulses, which are further conditioned using designed analog circuitry. The system architecture and measurement illustration are depicted in fig 2. Since it's an image free measurement system, a real time low frame rate (50 Hz) visual feedback was provided to the operator by processing the subset of the acquired frames. There is also a signal quality metric indicating the quality of vessel walls within the echoes for reliable acquisition. The acquisition system offered a temporal resolution of 2 ms at 500 Hz frame rate. The sampling rate of the system was 200 MHz.

High fidelity JVP signals were obtained by tracking the IJV vessel wall echoes. The acquired JVP signals were processed offline, and they were segmented into individual JVP cycles using the developed cycle segmentation algorithm. In the fiduciary point approach, vPTT was estimated as the numerical difference between time of occurrence of fiduciary points in the simultaneous JVP cycles. A typical JVP waveform comprises of six fiduciary points, three peaks (a, c, v) and three valleys (x, x', y) as indicated in fig 3 (e). Due to the morphological variations in JVP with various cardiovascular anomalies, the most frequent fiduciary points are to be considered. It had been observed that the critical peaks 'c' and 'v' are prominent in all participants as indicated in the earlier literatures [8], [9]. The fiduciary point identification algorithm was utilized to identify the critical points 'c' and 'v' of the simultaneous JVP cycles, and the fiduciary point-based PTTs were calculated. The time difference of the occurrence of JVP peaks c and v in simultaneous JVP cycles are estimated as vPTT<sub>c</sub> and vPTT<sub>v</sub> respectively.

A subset of individual JVP cycles were identified and cross correlation was implemented to estimate the vPTT<sub>|Xcorr</sub>. The cross-correlation function evaluates the time taken by a range of points in the venous pulse by comparing the JVP signals from the distal and proximal sites. An equal time window segment (T) of the cardiac cycle is selected for the cross-correlation technique [8]. The cross-correlation function is calculated as:

$$\mathcal{R}_{Xp, Xd}(\tau) = \frac{1}{\tau} \int_{-T/2}^{T/2} P(t; Xd) P(T + \tau; Xp) dt \quad (2)$$

Where  $P(t; Xp)$  is the group of points in the proximal and  $P(t; Xd)$  distal JVP signals respectively. The time shift at which the highest correlation between proximal and distal pulses occurs is identified as the vPTT<sub>|Xcorr</sub>. An alternative vPWV estimation is based on thresholding principle. The threshold was selected at 50% of the amplitude of the normalized cycle and the intersection of the proximal and distal pulses at the threshold line was automatically identified. The numerical shift between the intersection points of the proximal and distal cycles were estimated as the vPTT<sub>|th</sub>.

### E. Statistical Analysis

In this study, the vPWV values were measured using fiduciary point (vPWV<sub>|c</sub>, vPWV<sub>|v</sub>), thresholding (vPWV<sub>|th</sub>) and correlation (vPWV<sub>|Xcorr</sub>) techniques and their performance were compared. The averaged values are presented as mean  $\pm$  standard deviation (SD) format. To investigate the similarity or difference between the groups, box-and-whisker diagrams were plotted using median and interquartile ranges. The beat-to-beat coefficient of variation percentage (% CoV) was calculated as the percentage of the ratio of standard deviation to the mean value. The t-Test (paired two sample for means) was estimated to evaluate the agreement between corresponding parameters.

## III. RESULTS AND DISCUSSION

### A. Subject Characteristics

The selected participants had their systolic, diastolic blood pressure measurements as  $118.90 \pm 11.30$  mmHg,  $69.63 \pm 8.90$  mmHg respectively. The mean heart rate of 78 bpm was observed. The mean age of the cohort was  $24.8 \pm 2.11$  years with height =  $170.09 \pm 6$  cm and weight =  $65.54 \pm 9.54$  kg. The lowest vPWV was observed in vPWV<sub>|th</sub>,  $1.28 \pm 0.42$  m/s and the highest in vPTT<sub>|c</sub>,  $1.82 \pm 0.72$  m/s. The vPWV ranges of vPWV<sub>|v</sub> and vPWV<sub>|Xcorr</sub> were  $1.54 \pm 0.78$  m/s and  $1.65 \pm 0.50$  m/s respectively.

### B. Reliability of Data Collection

High fidelity JV echo frames (SNR > 20 dB) from the two proximally spaced ultrasound transducers were continuously tracked to generate corresponding JVP signals. Fig 3 (a) and (b) represent sharp IJV wall echoes from proximal and distal sites respectively. The JVP signals by processing corresponding echo frames are depicted in fig 3 (c) and (d). The acquisition system operates at a pulse repetition rate of 500 Hz, to ensure high fidelity reliable JVP cycles. Reliable signal acquisition was ensured by monitoring the SNR of echo frames > 20 dB and the signal quality score (SQ) to screen the near and far wall echoes, where the software automatically rejects frames of SQ < 20 % [9].

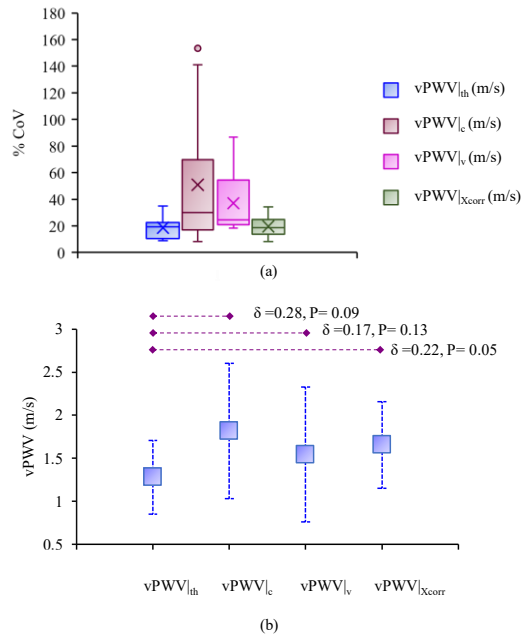


Fig 4. (a) Mean and standard deviation plot of the vPWV measurements (b) Box-Whisker plots of % CoV ranges in transit time based vPWV measurements

### C. Comparison Between the PWV Metrics

The cycle segmentation algorithm reliably segments the simultaneous JVP cycles obtained from the dual ultrasound system (beat-to-beat variability < 4%) to compute beat-to-beat vPTT and vPWV. Fig 4 (a) indicates the box-whisker comparison of the % CoV of vPWV calculated by different transit time approaches. The least % CoV in vPWV measurements was observed in vPWV<sub>th</sub>  $18.62 \pm 7.95$  and was found to be the most reliable vPWV metric. The mean and standard deviation of various vPWV measurements in comparison with vPWV<sub>th</sub> is indicated in fig 4 (b). The maximum deviation from mean PWV value was observed for vPWV<sub>c</sub> and the least for vPWV<sub>v</sub> measurements. The minimum variation in % CoV was observed in vPWV<sub>th</sub>  $18.62 \pm 7.95$ , while the vPWV<sub>c</sub>  $50.85 \pm 50.52$  had the maximum variation. The variations in vPWV<sub>Xcorr</sub> and vPWV<sub>v</sub> were  $37.93 \pm 33.16$  and  $19.71 \pm 7.10$  respectively. The lower ranges of the vPWV values resulted in the augmented % CoV values.

From the results, it was observed that the vPWV computed from fiduciary point 'c' had higher variability owing to poor repeatability in measurements. Also, vPWV<sub>c</sub> had the highest mean deviation in contrast with the thresholding technique. The vPWV<sub>v</sub> offered comparable vPWV ranges with the vPWV<sub>th</sub> values even though the % CoV ranges were high. While the cross correlation-based approach, offered highest beat-to-beat repeatability and had comparable % CoV as the vPWV<sub>th</sub>, the vPWV values were overestimated by 0.2 m/s. This variation in vPWV values was expected as the thresholding technique relies on two points, where the threshold line intersects with the two sections of the JVP, while the cross-correlation was dependent on the entire section of the pulse. But in this study the variation was non-significant. The thresholding and cross correlation metrics

offered a better performance in comparison with the fiduciary point approaches for vPWV estimation.

### D. Limitations

Even though all the fiduciary points were considered initially for PWV estimation, some fiduciary points were absent due to morphological variations in JVP across the individuals. However, the peaks 'c' and 'v' were consistently evident in all participants. The findings presented here are based on noninvasive measurements from local venous sites, since it is challenging to compare these against invasive vPWV measurements on humans until in intensive care units. Our further efforts include animal studies for validation of non-invasive vPWV metrics against invasive methods and comparison against other standard health markers [10], [11].

## IV. CONCLUSION

In this study, we have compared various vPWV computation metrics in terms of their repeatability and concurrence with the earlier reported ranges. The measured vPWV values, 1.54 to 1.82 m/s were within the specified ranges as in literature [4]. The study findings provide an insight into the viability of various transit time based local vPWV estimation techniques. The developed dual channel system could acquire reliable JVP signals for local vPWV estimation. The study reveals that the cross-correlation and thresholding metrics provided better estimation in comparison with fiduciary point approaches for local vPWV assessment.

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